Psychometric report for Statistical Reasoning

Report generated on 07-03-2026 by Koen Derks

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1. Summary

This section provides an overview of the assessment results and key characteristics of the score distribution.

1.1 Descriptive Statistics

The assessment included 128 participants. The table below summarizes key descriptive statistics of their achieved scores.

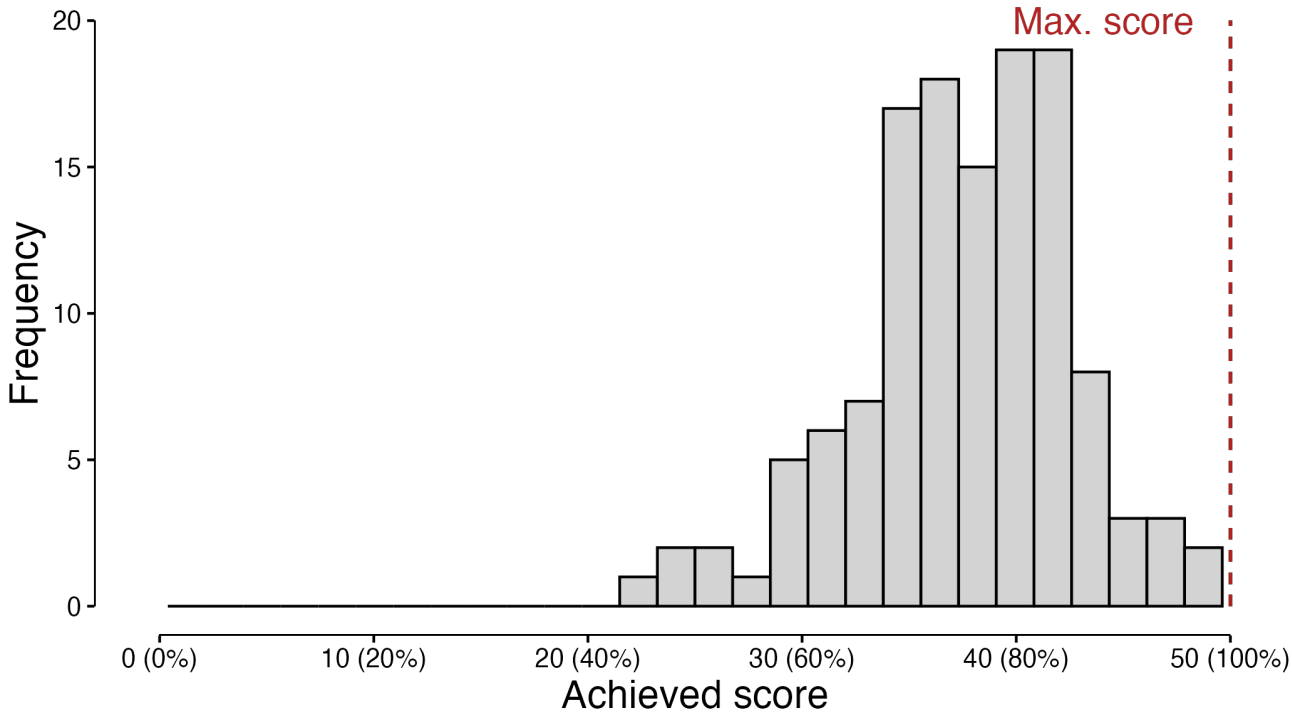
- The average score of 37.566 (75.132%) represents the central tendency of the scores and reflects the typical performance level.
- The median of 38.5 (77%) is the middle score when all participants are ordered from lowest to highest.
- The standard deviation of 5.038 (10.077%) measures the spread of scores around the average. Higher values indicate greater variability, while lower values indicate scores clustered near the mean.
- The skewness of -0.506 indicates that the distribution is negatively skewed, with a tendency toward higher scores. Positive skew reflects more low scores and fewer high scores, while negative skew indicates the opposite.
- The kurtosis of 0.239 means the distribution is approximately normal in peakedness, indicating a balanced spread of scores around the mean. Higher values indicate a stronger concentration of scores near the mean with more extreme values, while lower values indicate a flatter distribution.

Statistic	Value
Number of participants	128
Maximum possible score	50 (100%)
Minimum achieved score	22.75 (45.5%)
Maximum achieved score	49 (98%)
Average achieved score	37.566 (75.132%)
Median achieved score	38.5 (77%)
Standard deviation	5.038 (10.077%)
Skewness	-0.506
Kurtosis	0.239

1.2 Distribution of Achieved Scores

The histogram below visualizes the distribution of participant scores. The horizontal axis represents score ranges, and the vertical axis indicates the number of participants within each range.

- Most participants scored in the upper range (>70%), indicating that this range was most effective in differentiating participant performance.
- Look out for peaks near the maximum or minimum possible scores, as this may indicate ceiling effects (many very high scores) or floor effects (many very low scores), which can impact the assessment's ability to distinguish between stronger and weaker participants.
- Multiple peaks in the distribution may indicate heterogeneous participant groups with different performance levels.



2. Classical Assessment Analysis

Classical assessment analysis uses statistical indicators to evaluate the quality of the assessment (items). These metrics should be interpreted in the context of the assessment purpose, participant population, and sample size.

2.1 Assessments Statistics

The table below presents key evaluation metrics for the overall assessment. Cell colors follow the thresholds defined in the *Guideline Assessment Analysis*.

- *Average P (Difficulty)*: A value near 0 indicates a very difficult assessment, whereas a value near 1 indicates a very easy one. An assessment with moderate difficulty (typically between 0.3 and 0.8) tends to provide the greatest discriminatory power.

- *Average RIT and RIR (Discrimination)*: These statistics measure the extent to which items differentiate between higher- and lower-performing participants. Values below 0.2 (red) indicate weak discrimination, values between 0.2 and 0.3 (orange) indicate moderate discrimination, and values above 0.3 (green) indicate strong discrimination.

- *Cronbach's alpha (Internal Consistency)*: This is the internal consistency of the assessment. A Value above 0.7 (green) generally indicates acceptable reliability. Lower values may suggest inconsistent items or items that contribute little to the measurement of the intended construct. A very high value (>0.9) may indicate redundancy among items.

Average P	Average RIT	Average RIR	Cronbach's alpha
0.737	0.309	0.233	0.728

2.2 Item Statistics

This table summarizes key evaluation metrics for each individual item. Cell colors follow the thresholds defined in the *Guideline Assessment Analysis*.

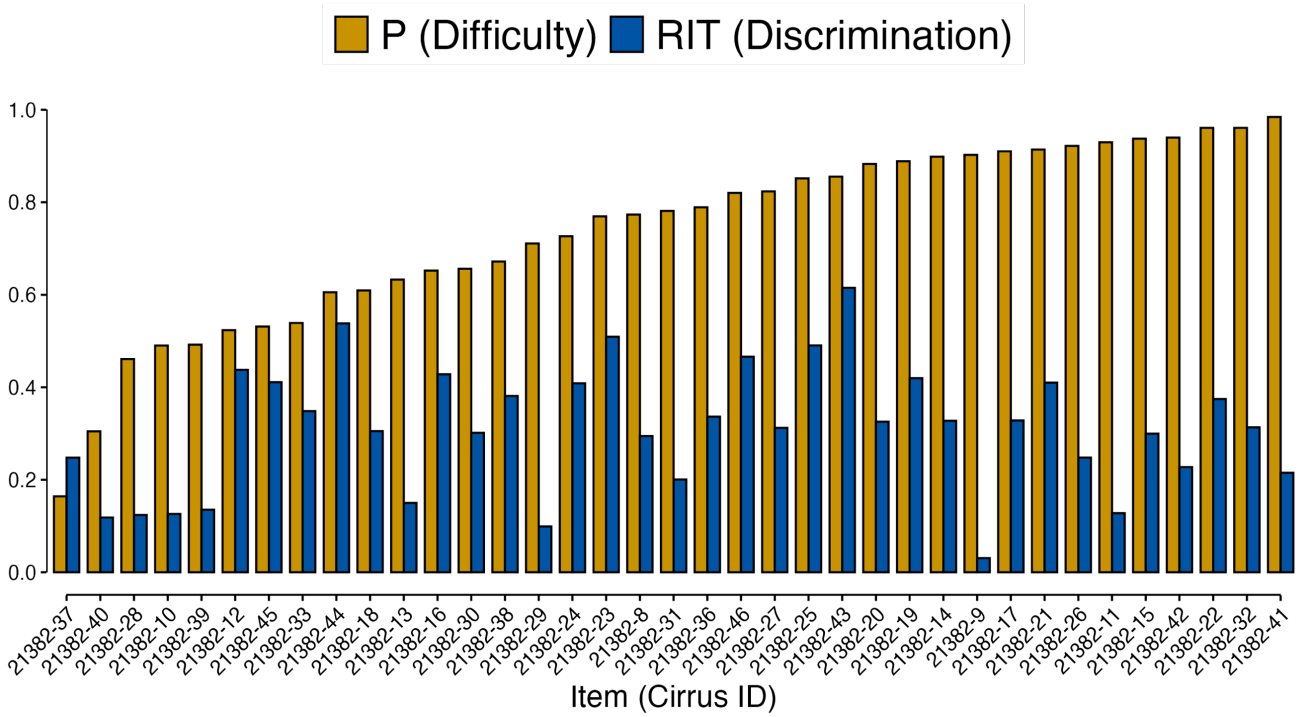
- *P (Item Difficulty)*: The P-value represents the proportion of participants who answered the item correctly. Values near 0 indicate very difficult items, while values near 1 indicate very easy items.
- *RIT (Item-Total Correlation)*: RIT represents the correlation between an item score and the total score. Higher values indicate that the item aligns well with overall performance. Negative values require immediate review, as they indicate that lower-performing participants answered the item correctly more often than higher-performing participants.
- *RIR (Item-Rest Correlation)*: RIR represents the correlation between an item and the total score excluding that item. Interpretation follows the same thresholds as RIT and provides a similar indication of item discrimination.
- *Alpha-if-deleted*: This statistic indicates the change in Cronbach's alpha if the item were removed from the assessment. An increase in alpha (red) suggests that the item may reduce internal consistency.

Item (Cirrus ID)	Mean	SD	P	RIT	RIR	Alpha-if-deleted
21382-8	2.320	0.790	0.773	0.295	0.143	0.734
21382-9	1.805	0.282	0.902	0.031	-0.025	0.732
21382-10	0.490	0.415	0.49	0.126	0.044	0.731
21382-11	0.930	0.257	0.93	0.128	0.077	0.728
21382-12	0.523	0.501	0.523	0.438	0.352	0.714
21382-13	0.633	0.484	0.633	0.15	0.054	0.732
21382-14	0.898	0.303	0.898	0.327	0.272	0.721
21382-15	0.938	0.243	0.938	0.3	0.255	0.723
21382-16	1.305	0.647	0.652	0.428	0.315	0.716
21382-17	0.910	0.165	0.91	0.328	0.298	0.723
21382-18	0.609	0.490	0.609	0.305	0.213	0.723
21382-19	1.777	0.429	0.889	0.42	0.346	0.716

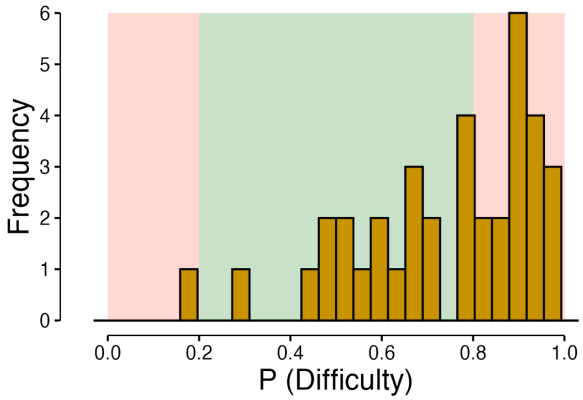
21382-20	0.883	0.323	0.883	0.325	0.266	0.721
21382-21	1.828	0.254	0.914	0.41	0.367	0.719
21382-22	1.922	0.238	0.961	0.374	0.333	0.721
21382-23	1.539	0.675	0.77	0.509	0.4	0.709
21382-24	0.727	0.447	0.727	0.409	0.331	0.716
21382-25	1.703	0.573	0.852	0.491	0.397	0.71
21382-26	0.922	0.269	0.922	0.248	0.197	0.724
21382-27	0.824	0.268	0.824	0.312	0.263	0.722
21382-28	0.461	0.500	0.461	0.124	0.025	0.734
21382-29	0.711	0.455	0.711	0.099	0.009	0.734
21382-30	0.656	0.477	0.656	0.301	0.212	0.723
21382-31	0.781	0.415	0.781	0.201	0.12	0.728
21382-32	0.961	0.195	0.961	0.313	0.278	0.723
21382-33	0.539	0.500	0.539	0.348	0.257	0.72
21382-36	0.789	0.410	0.789	0.337	0.262	0.72
21382-37	0.164	0.372	0.164	0.248	0.177	0.725
21382-38	0.672	0.471	0.672	0.381	0.297	0.718
21382-39	0.492	0.502	0.492	0.135	0.036	0.734
21382-40	0.305	0.462	0.305	0.118	0.027	0.733
21382-41	0.984	0.125	0.984	0.215	0.191	0.726
21382-42	0.940	0.199	0.94	0.227	0.189	0.725
21382-43	1.711	0.424	0.855	0.615	0.558	0.705
21382-44	1.211	0.695	0.605	0.538	0.429	0.706
21382-45	1.062	0.707	0.531	0.411	0.285	0.719
21382-46	1.641	0.498	0.82	0.466	0.383	0.712

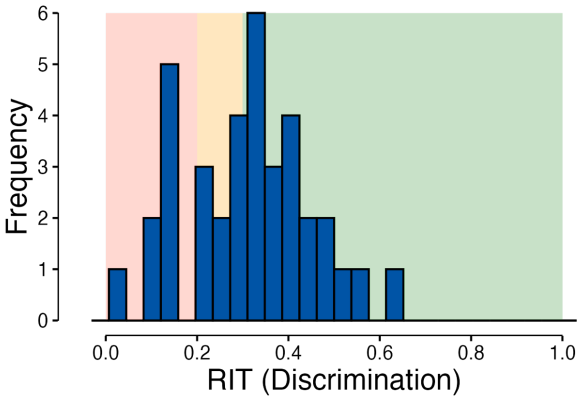
The plot below displays item difficulty (P-value) in relation to item discrimination (RIT).

- Items that are very easy or very difficult and show low discrimination may provide limited measurement information and may warrant revision.
- Items with moderate difficulty and strong discrimination generally contribute most effectively to reliable measurement.



The item difficulty (P-value) and item discrimination (RIT) are shown again in the figures below, with colored thresholds for quality assessment.

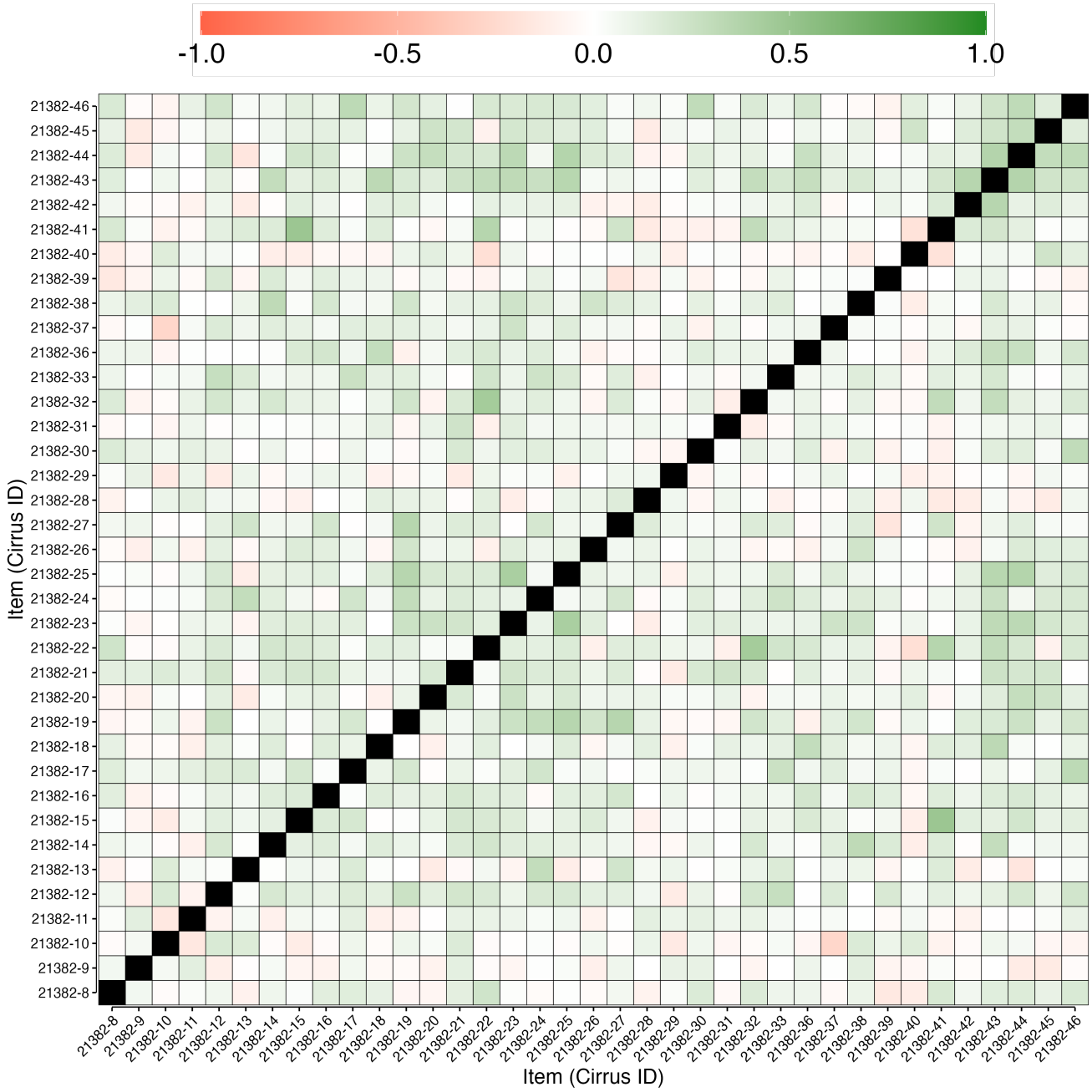




2.3 Item Correlations

This heatmap below shows correlations between all assessment items.

- Items showing strong positive correlations (>0.6) may measure very similar content and should be reviewed for potential redundancy.
- Items with negative correlations with multiple other items should be reviewed for potential scoring errors or content misalignment.



The table below below shows the five strongest positive and five strongest negative correlations. Red values indicate extremely strong correlations (>0.6) and negative correlations.

Item (Cirrus ID)	Item (Cirrus ID)	Correlation
21382-15	21382-41	0.488
21382-22	21382-32	0.443
21382-23	21382-25	0.417
21382-43	21382-44	0.369
21382-25	21382-44	0.366
21382-10	21382-37	-0.27
21382-22	21382-40	-0.211
21382-40	21382-41	-0.19
21382-27	21382-39	-0.167
21382-13	21382-44	-0.166

3. Flagged Items for Review

The table below summarizes items that may require review due to psychometric concerns. Each flagged item is listed with the specific issue(s) detected and an explanation for why it may warrant attention. The table is ordered so that items with the greatest number of issues are shown first.

Item	Issue	Explanation
21382-9	Very easy item; Low discrimination; Reduces test reliability; Negative item-rest correlation	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. Item-total correlation is below 0.20, indicating weak ability to distinguish between stronger and weaker participants. Cronbach's alpha increases when this item is removed, suggesting the item may not align well with the construct measured by the assessment. The item correlates negatively with the rest of the assessment, suggesting potential scoring errors or content misalignment.
21382-10	Low discrimination; Reduces test reliability	Item-total correlation is below 0.20, indicating weak ability to distinguish between stronger and weaker participants. Cronbach's alpha increases when this item is removed, suggesting the item may not align well with the construct measured by the assessment.
21382-11	Very easy item; Low discrimination	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. Item-total correlation is below 0.20, indicating weak ability to distinguish between stronger and weaker participants.

21382-13	Low discrimination; Reduces test reliability	Item-total correlation is below 0.20, indicating weak ability to distinguish between stronger and weaker participants. Cronbach's alpha increases when this item is removed, suggesting the item may not align well with the construct measured by the assessment.
21382-28	Low discrimination; Reduces test reliability	Item-total correlation is below 0.20, indicating weak ability to distinguish between stronger and weaker participants. Cronbach's alpha increases when this item is removed, suggesting the item may not align well with the construct measured by the assessment.
21382-29	Low discrimination; Reduces test reliability	Item-total correlation is below 0.20, indicating weak ability to distinguish between stronger and weaker participants. Cronbach's alpha increases when this item is removed, suggesting the item may not align well with the construct measured by the assessment.
21382-39	Low discrimination; Reduces test reliability	Item-total correlation is below 0.20, indicating weak ability to distinguish between stronger and weaker participants. Cronbach's alpha increases when this item is removed, suggesting the item may not align well with the construct measured by the assessment.
21382-40	Low discrimination; Reduces test reliability	Item-total correlation is below 0.20, indicating weak ability to distinguish between stronger and weaker participants. Cronbach's alpha increases when this item is removed, suggesting the item may not align well with the construct measured by the assessment.
21382-8	Reduces test reliability	Cronbach's alpha increases when this item is removed, suggesting the item may not align well with the construct measured by the assessment.
21382-14	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.
21382-15	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.
21382-17	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.
21382-19	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.
21382-20	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between

		participants.
21382-21	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.
21382-22	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.
21382-25	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.
21382-26	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.
21382-27	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.
21382-32	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.
21382-37	Very difficult item	Less than 20% of participants answered correctly, suggesting the item may be ambiguous, miskeyed, or cover material not sufficiently taught.
21382-41	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.
21382-42	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.
21382-43	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.
21382-46	Very easy item	More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants.